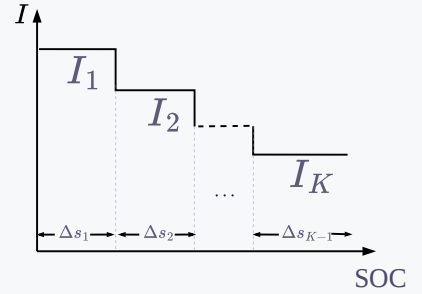


DECISION VARIABLES

$$\boldsymbol{\theta} = [I_1, \dots, I_K, \Delta s_1, \dots, \Delta s_{K-1}]^\top$$

$$\boldsymbol{\theta} \in \Theta_{\text{proto}}$$

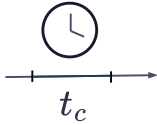
$$\Delta s_K = s_f - s_0 - \sum_{k=1}^{K-1} \Delta s_k$$



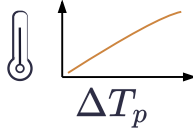
propose protocol

OBJECTIVES (ALL MINIMIZED)

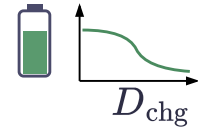
$$\min_{\boldsymbol{\theta}} f(\boldsymbol{\theta}) = [t_c(\boldsymbol{\theta}), \Delta T_p(\boldsymbol{\theta}), D_{\text{chg}}(\boldsymbol{\theta})]^\top$$



charging time



peak temperature rise



degradation score

evaluate objectives

CONSTRAINTS

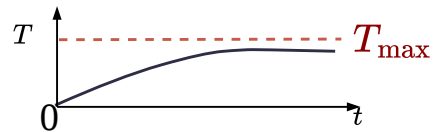
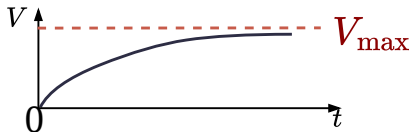
$$V_{\text{peak}}(\boldsymbol{\theta}) \leq V_{\text{max}}$$

(path voltage limit)

$$T_{\text{peak}}(\boldsymbol{\theta}) \leq T_{\text{max}}$$

(path temperature limit)

$$I_{\min} \leq I_k \leq I_{\max}, \quad \sum_k \Delta s_k = s_f - s_0, \quad \Delta s_K \geq \Delta s_K^{\min}$$



constrained Pareto search